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Machine Learning in the Chinese Stock Market

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1 Big picture

- **Question.** How much can machine learning improve stock-return prediction in the Chinese A-share market, which predictors matter most, and do those forecasts remain economically useful after market frictions such as transaction costs, short-sale constraints, and daily price limits?
- **Setting.** The paper studies more than **3,900 A-share stocks** from **January 2000 to June 2020**, with an out-of-sample test period running from **2012 to 2020**. The market is institutionally unusual because it is heavily retail-dominated, contains many state-owned enterprises (SOEs), and historically has tight short-selling constraints.
- **Core empirical challenge.** High-dimensional return prediction is especially vulnerable to overfitting. The paper therefore needs a design that can compare many flexible models on genuinely unseen data, rather than on in-sample fit.
- **Main conceptual contribution.** The paper ports the modern Gu-Kelly-Xiu machine-learning asset-pricing framework to China, but also expands the predictor set with China-specific variables and studies whether the Chinese market's institutional features change *which* signals matter and *where* predictability comes from.
- **Main predictor design.** The authors build a broad signal set with **1,160 covariates** in the core predictor vector: **90 stock characteristics**, **11 macroeconomic variables**, all characteristic-macro interactions, and **80 industry dummies**. The broader data collection contains **94 stock-level characteristics**, including China-specific variables and ownership indicators used in subgroup work.
- **Main model comparison.** The paper compares ordinary least squares, a simple three-variable OLS benchmark, penalized regressions, tree methods, VASA, and neural networks with one to five hidden layers. Hyperparameters are tuned on a validation sample, and performance is judged on a rolling out-of-sample basis.
- **Main monthly result.** Machine learning methods strongly outperform linear benchmarks at the monthly horizon. On the *full sample*, the highest monthly out-of-sample R^2 is **2.71%** for GBRT, while neural networks deliver around **2.0%–2.6%**. On the *bottom 30% of stocks by market capitalization*, predictability becomes much stronger: GBRT reaches **7.27%**, NN4 reaches **5.50%**, and NN5 reaches **5.33%**.
- **Key heterogeneity result.** Monthly predictability is much larger for **small stocks**, **small-shareholder stocks**, and **non-SOEs**, all groups in which retail trading and speculative activity are likely to be more important. At the annual horizon, the pattern flips: **large stocks** and especially **SOEs** become more predictable.
- **Key model-selection result.** Although several flexible methods score well on raw out-of-sample R^2 , neural networks look strongest once the paper applies the unconditional SPA test and the **conditional superior predictive ability (CSPA)** test. In Table 5, NN4 has only **2** total CSPA rejections, the lowest among all models.
- **Key variable-importance result.** In contrast to the U.S. evidence in Gu et al. (2020), the most important Chinese predictors are **liquidity-related signals**. Fundamental variables

come second, while classic momentum variables are not the dominant force overall. China-specific predictors such as **abnormal turnover ratio** and the **trend factor** also matter a lot.

- **Main portfolio result.** Predictability translates into large economic gains. In the full-sample value-weighted decile strategies, the best long-short Sharpe ratio is **3.45** (NN3), and the best long-only Sharpe ratio is **1.76** (NN3). Neural networks and VASA also hold up well during the **2015 Chinese stock-market crash**.
- **Main credibility result.** The paper does not stop at statistical fit. It shows that the portfolio gains remain meaningful after **transaction costs**, after removing **small stocks**, and after accounting for **daily price limits**. For NN4, the long-only Sharpe ratio falls only from **1.68** to **1.46** when imposing a conservative **80 bps** transaction cost, and from **1.78** to **1.70** when excluding stocks locked at price limits.
- **Economic takeaway.** The Chinese stock market is not just “the U.S. market with different coefficients.” Its predictability structure is different. Retail-dominated trading raises short-horizon predictability, while the state’s prominent role appears to matter more at longer horizons, especially for large firms and SOEs.

2 Data and measurement (institutional context and key objects)

2.1 Why the Chinese stock market is the right laboratory

- The paper emphasizes **three institutional features**.
 - **Retail dominance.** In the 2019 Shanghai Stock Exchange yearbook, China has about **214.5 million investors**, of which **213.8 million** are individuals and only **0.7 million** are institutions. Individuals hold **99.8%** of all accounts holding stocks.
 - **State influence.** SOEs remain central in China’s capital markets, and government policy shapes IPO timing, refinancing, and other financing decisions.
 - **Short-sale restrictions.** Broad, liquid long-short implementation is much harder than in the U.S., so long-only evidence is especially important.
- Share turnover is also unusually high: the value of shares traded stands at about **224%** of market capitalization in 2019, versus **108%** in the U.S.

Interpretation. These features make China a useful setting to study whether stock-return predictability is driven more by speculative trading and institutional frictions than by the mix of signals that dominates in developed markets.

2.2 Data source and sample construction

- Daily and monthly total returns for all A-share stocks come from the **Wind Database**.

- Quarterly financial statement data come from **CSMAR**.
- The one-year Chinese government bond yield from CSMAR proxies for the risk-free rate used to form excess returns.
- The full sample spans **January 2000 to June 2020**.
- The test period used for out-of-sample evaluation runs from **2012 to 2020**.
- The paper studies more than **3,900 listed A-share firms**.

Interpretation. The authors work with a very large panel of returns and accounting data, which is important because flexible machine-learning methods require a lot of observations to avoid purely noisy complexity.

2.3 Stock-level characteristics

- The paper collects **94 stock characteristics** in total.
- Of these, **86** follow definitions from Green et al. (2017), **4** are China-specific characteristics from earlier Chinese asset-pricing papers, and **4** are binary ownership variables used mainly in subgroup analysis.
- In the main predictor vector, the paper uses **90 stock characteristics**; the ownership variables are treated separately in many specifications.
- Characteristics are updated at mixed frequencies:
 - **22** monthly,
 - **51** quarterly,
 - **6** semi-annually,
 - **15** annually.
- To reduce the influence of outliers, each continuous characteristic is ranked cross-sectionally each period and mapped into the interval $[-1, 1]$.

Interpretation. The paper follows the modern “many characteristics” approach from the U.S. literature, but modifies it for Chinese institutions and data availability.

2.4 Macroeconomic predictors

The paper augments firm-level characteristics with **11 macroeconomic variables**:

- dividend-price ratio (dp),
- dividend payout ratio (de),
- book-to-market ratio (bm),
- stock variance ($svar$),

- earnings-price ratio (*ep*),
- net equity issuance (*ntis*),
- term spread (*tms*),
- inflation (*infl*),
- monthly turnover (*mtr*),
- M2 growth (*m2gr*),
- international trade growth (*itgr*).

Interpretation. These variables let the expected-return function vary with the macro environment, not just with firm characteristics.

2.5 Industry controls and the total predictor dimension

- The paper adds **80 industry dummies**, based on the 2012 China Securities Regulatory Commission industry classification.
- Combining firm characteristics, macro variables, and interactions yields a high-dimensional predictor vector with **1,160** total covariates in the core design.

Interpretation. The paper deliberately creates a difficult prediction problem: lots of signals, lots of possible nonlinearities, and lots of room for overfitting unless the design is disciplined.

2.6 China-specific stock characteristics

Two China-specific signals receive especially strong attention in the paper:

- **Abnormal turnover ratio (*atr*)**. Introduced by Pan et al. (2015), this is designed to isolate speculative trading from more ordinary liquidity effects.
- **Trend factor (*er_trend*)**. From Liu et al. (2020), this is meant to capture persistent price-and-volume trends associated with market sentiment and noise-trader demand.

Interpretation. These variables are designed to capture behavior that is plausibly much more important in a retail-driven market than in a more institutional one.

2.7 Subsamples used throughout the paper

The paper repeatedly splits the sample in three ways.

- **By market capitalization.**
 - Top **70%** of firms by monthly market equity.
 - Bottom **30%** of firms by monthly market equity.

- **By average market capitalization per shareholder.**

$$A.M.C.P.S. = \frac{\text{Market Cap}}{\text{Number of Shareholders}}. \quad (1)$$

- Top **70%** by A.M.C.P.S.
- Bottom **30%** by A.M.C.P.S.

- **By ownership type.**

- SOEs,
- non-SOEs.

Interpretation. The A.M.C.P.S. split is the paper’s proxy for the prevalence of small retail shareholders. Low A.M.C.P.S. means a given market value is spread across many small holders.

2.8 Main outcome variables

The paper studies several related outcome objects.

- **One-month-ahead excess stock return**, $r_{i,t+1}$.
- **Out-of-sample predictive R^2** , the main statistical metric.
- **Variable importance measures**, based on the drop in predictive R^2 when a predictor is shut down.
- **Portfolio performance metrics**, especially average return, Sharpe ratio, maximum draw-down, and the most negative monthly loss.

Interpretation. The paper does not rely on a single score. It checks whether a model forecasts returns well statistically, whether it uses sensible predictors, and whether the resulting signals survive real trading frictions.

3 Models and empirical specifications (all equations + interpretation)

3.1 General prediction setup

The paper starts from the additive prediction-error decomposition

$$r_{i,t+1} = E_t[r_{i,t+1}] + \varepsilon_{i,t+1}, \quad (2)$$

and assumes

$$E_t[r_{i,t+1}] = g(z_{i,t}), \quad (3)$$

where $z_{i,t}$ is a large predictor vector and $g(\cdot)$ is left unspecified.

Interpretation. The whole exercise is to estimate a flexible conditional expectation function for excess returns without imposing a linear structure up front.

3.2 The core predictor vector

The paper writes the predictor vector as

$$z_{i,t} = (c'_{i,t}, x'_t \otimes c'_{i,t}, d'_{i,t})', \quad (4)$$

where

- $c_{i,t}$ is a 90×1 vector of stock characteristics,
- x_t is a 11×1 vector of macro predictors,
- $d_{i,t}$ is a vector of **80 industry dummies**,
- $\otimes$ denotes the Kronecker product.

This yields

$$90 \times (11 + 1) + 80 = 1,160 \quad (5)$$

predictors in the core model.

Interpretation. The interaction term allows the effect of a stock characteristic to vary with macro conditions. For example, a liquidity measure can have a different relation to expected returns in a high-inflation state than in a low-inflation state.

3.3 Model classes

The paper compares **13 models**:

- **OLS**,
- **OLS-3** using only size, book-to-market, and momentum,
- **PLS**,
- **LASSO**,
- **elastic net**,
- **GBRT**,
- **random forest**,
- **VASA**,
- **NN1** through **NN5**.

The paper uses a **Huber loss** variant for OLS, OLS-3, LASSO, elastic net, and GBRT to reduce sensitivity to extreme returns; the Huber tuning parameter is set to $M = 1.35$.

Interpretation. The comparison spans the full ladder from simple linear models to highly flexible nonlinear methods.

3.4 Training, validation, and testing design

The sample is split into three temporally ordered blocks:

- **training:** 2000–2008,
- **validation:** 2009–2011,
- **testing:** the next 12 months, beginning in 2012.

The paper then uses an annual rolling scheme:

- when the model is refit, the training window expands by one year;
- the validation window rolls forward by one year but keeps the same length;
- the next 12 months form the new test period.

Interpretation. This is the core safeguard against overfitting: the model is always evaluated on future observations that were never used for estimation or tuning.

3.5 Main performance metric

For model S , the paper defines non-demeaned out-of-sample predictive R^2 as

$$R_{\text{OOS},S}^2 = 1 - \frac{\sum_{(i,t) \in T} \left(r_{i,t} - \hat{r}_{i,t}^{(S)} \right)^2}{\sum_{(i,t) \in T} r_{i,t}^2}, \quad (6)$$

where T is the set of test-sample observations.

Interpretation. A positive R_{OOS}^2 means the model beats the naive forecast of zero excess return. The larger the number, the more useful the forecasts are in genuinely unseen data.

3.6 Variable-importance measure

For each predictor and each model, variable importance is computed from the drop in predictive R^2 when that predictor is set to zero in the training sample, then averaged across periods.

Interpretation. This does not say a variable is “causal.” It says the model loses forecasting power when that variable is neutralized.

3.7 Model-selection tests beyond raw R^2

The paper uses:

- the **unconditional superior predictive ability (USPA)** test of Hansen (2005),
- the **conditional superior predictive ability (CSPA)** test of Li et al. (2020).

For the CSPA tests, the paper conditions on six macro state variables:

- inflation (*infl*),
- M2 growth (*m2gr*),
- aggregate book-to-market (*bm*),
- dividend-price ratio (*dp*),
- market turnover (*mtr*),
- market stock variance (*svar*).

Interpretation. Two models can have similar overall R^2 but behave differently across macro states. CSPA is meant to reveal which methods are more robust when the environment changes.

3.8 Portfolio construction

At the end of each month:

- the paper forms one-month-ahead predicted returns for all stocks,
- sorts stocks into deciles based on the predictions,
- constructs value-weighted portfolios.

It studies two main strategies:

- **long-short:** buy decile 10, short decile 1;
- **long-only:** hold only decile 10.

Benchmarks include:

- the **1/N** equally weighted portfolio,
- the **CSI 300** market index in cumulative-return figures.

Interpretation. The long-short strategy is a clean way to measure signal strength, while the long-only strategy is more realistic in a market with tight short-selling constraints.

3.9 Transaction-cost and trading-friction analysis

The paper studies three main costs:

- **commission,**
- **stamp tax,**
- **slippage.**

Slippage is estimated using:

- a **TWAP** over the first 30 minutes of the first trading day of the month,
- a **VWAP** that uses imputed minute-level volume weights.

The paper then applies conservative round-trip cost assumptions of **20, 40, 60, and 80 bps**.

Interpretation. This section asks whether the machine-learning gains are merely paper profits or whether they survive plausible implementation costs.

3.10 Daily price-limit implementation check

On rebalancing dates:

- buy targets that are closed at the **upper price limit** are excluded,
- sell targets that are closed at the **lower price limit** are postponed until they become tradable.

Interpretation. This is a China-specific implementation check. It matters because daily price limits can prevent an investor from actually trading on the model's signals.

4 Identification and empirical design (what makes the predictive claims credible)

4.1 Why this paper is about predictive credibility, not causal identification

- The paper is not trying to estimate the causal effect of one specific variable on returns.
- Instead, it asks whether a large set of signals can forecast returns out of sample, and which model class does that most reliably.

Interpretation. “Identification” here means credible prediction design: strict time ordering, honest evaluation on unseen data, and robustness to trading frictions.

4.2 How the paper guards against overfitting

- Estimation, hyperparameter tuning, and evaluation are done on **separate samples**.
- Test data are never used during model fitting.
- Models are re-estimated only annually, which reduces computational overfitting and keeps the design close to feasible implementation.

Interpretation. A model can fit historical returns extremely well just by memorizing noise. This train/validate/test split is the paper's main defense against that problem.

4.3 Why the benchmarks matter

- The paper includes very simple benchmarks such as OLS and OLS-3.
- If complex models only barely beat these parsimonious alternatives, the extra complexity would be hard to justify.

Interpretation. The large performance gaps relative to OLS-3 are part of the paper’s credibility story: complexity appears to buy something real.

4.4 Why the paper does not rely on one single statistic

- The authors look at monthly and annual R^2 .
- They compare many subsamples.
- They apply USPA and CSPA tests.
- They convert forecasts into portfolio performance.
- They then add realistic frictions such as transaction costs and daily price limits.

Interpretation. Strong evidence should survive multiple lenses. This paper deliberately checks the models from several angles.

4.5 Why the subgroup analysis matters

- If predictability were equally strong everywhere, the paper would say only that machine learning helps in general.
- Instead, the pattern is informative:
 - short-horizon predictability is strongest for small, retail-heavy segments;
 - long-horizon predictability is strongest for large firms and SOEs.

Interpretation. These systematic differences help connect the statistical results to economic features of the Chinese market.

4.6 What the paper can and cannot say

- The paper can say that certain variables and models improve out-of-sample forecasts.
- It can also say that the Chinese market’s predictability pattern differs from the U.S.
- It cannot cleanly separate **risk premia**, **mispricing**, and **institutional frictions** using this reduced-form design alone.
- The retail-intensity proxy A.M.C.P.S. is useful but not perfect, because the paper does not directly observe investor type for every stock.

Interpretation. The economic interpretation is plausible and well-argued, but it is still an interpretation layered on top of a predictive design.

4.7 Relation to the U.S. machine-learning asset-pricing literature

- The paper explicitly benchmarks itself against **Gu et al. (2020)** for the U.S.
- The Chinese market produces much higher short-horizon predictability for small stocks.
- The ranking of predictors also changes: liquidity is central in China, whereas conventional price-trend indicators play a larger role in the U.S. evidence.

Interpretation. The paper’s contribution is not only “machine learning works in China,” but also “machine learning reveals that China is predictably different from the U.S.”

5 Tables and figures

5.1 Table 1: monthly out-of-sample predictive R^2

Read.

- On the **full sample**, OLS achieves only **0.81%**, while GBRT reaches **2.71%**; NN1–NN5 range from about **2.04%** to **2.58%**.
- On the **top 70%** largest stocks, predictability becomes weak. Several models even turn negative: OLS is **-0.89%**, GBRT is **-0.38%**, and RF is **-0.04%**.
- On the **bottom 30%** smallest stocks, predictability becomes very strong: GBRT reaches **7.27%**, RF **6.10%**, NN4 **5.50%**, and NN5 **5.33%**.
- For the **bottom 30% by A.M.C.P.S.**, some models again perform very well: PLS reaches **7.08%**, RF **3.93%**, NN4 **3.56%**, and NN5 **3.67%**.
- For **SOEs**, monthly predictability is modest for most non-neural models, but the neural networks remain positive, with NN5 at **1.68%**.
- For **non-SOEs**, monthly predictability is much stronger, with GBRT at **3.67%**, RF at **3.02%**, and neural networks around **2.35%–2.92%**.

Economic message. Monthly predictability in China is not evenly distributed. It is concentrated in smaller, more retail-heavy, and less state-dominated parts of the market.

Table 1

Monthly out-of-sample predictive R^2 in percentage. This table reports monthly out-of-sample predictive R^2 of forecast models for different subgroups of firms: (1) the full sample; (2) the sample excluding firms with bottom 30% market values; (3) the sample including only the firms with the 30% bottom market values; (4) the sample including firms with top 70% average market capitalization per shareholder; (5) the sample including only the firms with the bottom 30% average market capitalization per shareholder; (6) state-owned-enterprises; and (7) non-state-owned-enterprises. The models considered include ordinary least squares (OLS) regression, OLS using only size, book-to-market and momentum (OLS-3), partial least squares regression (PLS), least absolute shrinkage and selection operator (LASSO), elastic net (Enet), gradient boosted regression trees (GBRT), random forest (RF), variable subsampling aggregation (VASA), and neural networks with 1 to 5 layers (NN1–NN5). “+H” indicates that the model is trained using Huber loss instead of l_2 loss. SOE and Non-SOE represent the subgroups of state-owned and non-state-owned enterprises, respectively. All the numbers are expressed as a percentage.

	OLS +H	OLS-3 +H	PLS +H	LASSO +H	Enet +H	GBRT +H	RF	VASA	NN1	NN2	NN3	NN4	NN5
All	0.81	0.77	1.28	1.43	1.42	2.71	2.44	1.37	2.07	2.04	2.28	2.49	2.58
Top 70%	-0.89	0.23	0.56	0.55	0.36	-0.38	-0.04	0.34	0.41	0.51	0.74	0.47	0.72
Bottom 30%	1.33	1.57	2.35	2.74	3.00	7.27	6.10	2.90	4.52	4.32	4.57	5.50	5.33
A.M.C.P.S. Top 70%	0.47	1.31	0.55	1.36	1.53	1.39	1.69	1.41	1.72	1.67	2.01	1.96	2.03
A.M.C.P.S. Bottom 30%	1.49	-0.31	7.08	1.12	1.22	1.48	3.93	1.29	2.78	2.79	2.84	3.56	3.67
SOE	-0.06	0.52	0.68	0.85	0.79	0.01	0.80	0.75	1.10	1.18	1.28	1.30	1.68
Non-SOE	1.12	0.87	1.50	1.64	1.65	3.67	3.02	1.80	2.41	2.35	2.64	2.92	2.90

Table 2

Annual out-of-sample predictive R^2 in percentage. This table reports annual out-of-sample predictive R^2 of forecast models for different subgroups of firms: (1) the full sample; (2) the sample excluding firms with bottom 30% market values; (3) the sample including only the firms with the 30% bottom market values; (4) the sample including firms with top 70% average market capitalization per shareholder; (5) the sample including only the firms with the bottom 30% average market capitalization per shareholder; (6) state-owned-enterprises; and (7) non-state-owned-enterprises. The models considered include ordinary least squares (OLS) regression, OLS using only size, book-to-market and momentum (OLS-3), partial least squares regression (PLS), least absolute shrinkage and selection operator (LASSO), elastic net (Enet), gradient boosted regression trees (GBRT), random forest (RF), variable subsampling aggregation (VASA), and neural networks with 1 to 5 layers (NN1–NN5). “+H” indicates that the model is trained using Huber loss instead of l_2 loss. SOE and Non-SOE represent the subgroups of state-owned and non-state-owned enterprises, respectively. All the numbers are expressed as a percentage.

	OLS +H	OLS-3 +H	PLS +H	LASSO +H	Enet +H	GBRT +H	RF	VASA	NN1	NN2	NN3	NN4	NN5
All	3.22	3.27	3.51	4.47	4.33	4.53	4.15	4.19	4.26	5.39	5.21	5.17	5.24
Top 70%	3.74	4.23	4.18	5.30	5.20	5.23	4.61	4.95	7.17	5.68	5.79	5.80	6.48
Bottom 30%	3.46	3.73	3.80	4.74	4.59	4.92	3.92	4.40	6.54	5.36	5.47	5.48	6.02
A.M.C.P.S. Top 70%	3.90	3.42	4.91	4.02	4.66	4.07	4.77	4.34	4.08	5.78	5.51	6.06	6.33
A.M.C.P.S. Bottom 30%	0.59	2.40	3.05	1.50	3.75	2.97	1.75	3.60	1.45	3.87	4.02	1.72	1.06
SOE	4.71	5.80	5.84	6.98	6.89	5.81	6.53	6.57	8.98	6.87	6.82	7.20	8.18
Non-SOE	3.08	3.12	3.09	4.10	3.99	4.77	3.22	3.80	5.88	4.87	5.07	4.87	5.32

Table 3
Average out-of-sample predictive R^2 in percentage for NN1 to NN5. This table reports the average out-of-sample predictive R^2 for the neural networks NN1 to NN5 for different subgroups of firms: (1) the sample including only the firms with the 30% bottom market values; (2) the sample excluding firms with bottom 30% market values; (3) the sample including the firms with the bottom 30% average market capitalization per shareholder; (4) the sample including firms with the top 70% average market capitalization per shareholder; (5) non-state-owned enterprises; (6) state-owned enterprises. In addition, we add the corresponding numbers for the top and bottom 1,000 companies for the US market as analyzed in Gu et al. (2020), their tables 1 and 2. All the numbers are expressed in percentage values. The numbers in parentheses are the average out-of-sample predictive R^2 for all models, excluding OLS.

	Bottom 30%	Top 70%	Small-shareholder	Large-shareholder	Non-SOE	SOE	US bottom	US top
Monthly	4.85(4.18)	0.57(0.37)	3.13(2.62)	1.88(1.55)	2.64(2.26)	1.31(0.91)	0.44(0.36)	0.62(0.41)
Annual	5.77(4.91)	6.18(5.39)	2.42(2.60)	5.73(4.95)	5.20(4.34)	7.61(6.87)	4.37(4.68)	4.30(3.34)

5.2 Table 2: annual out-of-sample predictive R^2

Read.

- At the **annual horizon**, all full-sample models improve sharply. The best full-sample annual R^2 is **5.39%** for NN2.
- The **top 70%** large-stock sample now becomes much more predictable: NN1 reaches **7.17%**, NN5 reaches **6.48%**, and even the linear models exceed **3.7%**.
- The **bottom 30%** small-stock sample still looks predictable, but it is no longer dominant. NN5 is **6.02%**, not much larger than large-stock performance.
- The most striking annual result is for **SOEs**: NN1 reaches **8.98%**, NN5 reaches **8.18%**, and the average annual neural-network R^2 for SOEs is **7.61%**.

Economic message. Retail trading seems to drive short-horizon predictability, while large firms and especially SOEs become easier to forecast over longer horizons.

5.3 Table 3: China versus U.S. predictability

Read.

- For the **bottom 30%** of Chinese stocks, the average monthly neural-network R^2 is **4.85%**, versus only **0.44%** for the bottom-1,000 U.S. firms in Gu et al. (2020).
- For the **top 70%** of Chinese stocks, the average monthly neural-network R^2 is only **0.57%**, close to the **0.62%** for top U.S. firms.
- At the **annual horizon**, Chinese large stocks average **6.18%**, above the U.S. top-firm benchmark of **4.30%**.
- Chinese **SOEs** are even more predictable annually, with an average neural-network R^2 of **7.61%**.

Economic message. The headline China-U.S. difference is not “China is always more predictable.” It is that China is *especially* more predictable where retail trading or state influence is most important.

5.4 Table 4 and Figure 1: macroeconomic variable importance

Read.

- Across models, **inflation** and **net equity issuance** (*ntis*) emerge as the most influential macro predictors.

Table 4
Relative variable importance of macroeconomic variables. This table reports the R^2 -based variable importance for macroeconomic variables in each model. For a given model, the sum of variable importance is normalized to one. All values are in percentage.

	PLS	LASSO	Enet	GBRT	RF	VASA	NN1	NN2	NN3	NN4	NN5
	-H	-H	-H	-H							
<i>dp</i>	0.00	8.65	4.07	9.11	9.44	1.34	2.17	2.96	3.31	4.01	1.63
<i>de</i>	0.00	1.06	1.78	9.40	8.59	1.32	5.46	5.86	5.28	6.57	5.78
<i>bm</i>	1.06	34.33	26.24	8.97	8.34	0.00	8.46	7.23	5.99	7.99	9.53
<i>svwr</i>	0.00	0.00	0.13	7.76	8.86	15.88	2.12	2.93	3.23	3.97	1.59
<i>ep</i>	0.00	0.68	0.98	8.09	9.86	46.41	2.14	2.94	3.21	3.99	1.59
<i>ntis</i>	41.19	14.54	14.37	12.30	9.12	0.00	18.35	18.78	20.01	16.36	17.60
<i>tms</i>	0.00	0.00	0.52	8.74	9.17	12.86	2.13	2.93	3.31	4.00	1.58
<i>infl</i>	21.14	21.86	28.63	9.11	11.92	0.00	40.61	38.41	38.16	31.97	38.12
<i>mtr</i>	0.00	0.00	0.26	9.22	10.22	22.19	2.12	2.95	3.28	4.00	1.58
<i>m2gr</i>	18.33	16.57	19.12	8.22	7.12	0.00	8.19	7.57	6.63	8.51	8.50
<i>itgr</i>	18.28	2.32	3.91	9.52	7.36	0.00	8.24	7.44	7.57	8.62	10.50

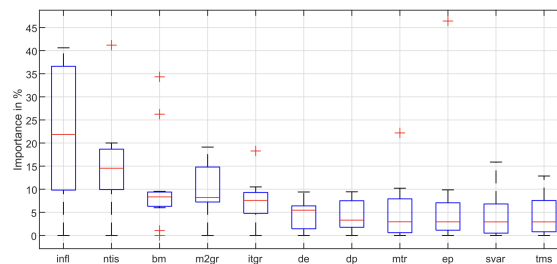


Fig. 1. Variable importance for eleven macroeconomic variables. This figure illustrates a box plot for the relative variable importance in Table 4 aggregated for each of the eleven macroeconomic variables.

- PLS heavily weights *ntis*, while LASSO and elastic net give large weight to **aggregate book-to-market**.
- GBRT and RF distribute weight more evenly across macro variables, consistent with their ability to detect more complicated interactions.

Economic message. Macro state variables matter, but their importance depends on model class. Neural networks and tree methods do not see the macro environment the same way as sparse linear models.

5.5 Figure 2: stock-characteristic importance

Read.

- The most influential stock-level predictors are mainly **liquidity and trading** measures:
 - `std.dolvol`,
 - `std.turn`,
 - `zerotrade`,
 - `ill`.
- Important **fundamental** variables also rank highly, including `chaotia`, `chempia`, `mve`, `nincr`, `chpmia`, and `bm_ia`.
- Risk-related variables such as `idiovol`, `volatility`, and `beta` also matter.
- The China-specific variables `atr` and `er_trend` are among the more important predictors.

Economic message. This is one of the paper's sharpest findings: China's return predictability is organized more around liquidity and speculative trading than around the classic momentum ranking from the U.S. literature.

5.6 Table 5: USPA and CSPA model-selection results

Read.

- OLS and OLS-3 are rejected heavily under both unconditional and conditional predictive-ability tests.



Fig. 2. Characteristic importance for all models. This figure shows the ordering of all stock-level characteristics ranked by their overall model contribution. Characteristics on the vertical axis are ordered based on the sum of their ranks over all models, with the most influential characteristics on the top and the least influential on the bottom. Columns correspond to the individual models, and the color gradients within each column indicate the most influential (dark blue) to the least influential (white) variables.

Table 5
Comparison of (un)conditional superior predictive ability based on full sample. The first column reports the number of rejections of the one-versus-one USPA test for row models at the 5% significance level based on the full sample. The next six columns report similar summary statistics of the conditional superior predictive ability tests (Li et al. (2020)) for different conditioning variables. For the CSPA tests, the entries report the number of rejections of the CSPA tests against the rest 12 competing models for a specific pair of the row model and the column conditioning variable. The last column reports the total number of rejections of the CSPA tests. For each entry, an asterisk indicates the rejection of a one-versus-all test at the 5% significance level.

	USPA	CSPA Test						Total
		infl	m2gr	bm	dp	mtr	svr	
OLS(+H)	10*	9*	11*	11*	10*	9	9	59
OLS-3(+H)	10*	8*	10*	9*	10*	9*	10*	56
PLS	3*	4*	5*	3	5*	6*	6	29
LASSO(+H)	3*	3	2	1	0	3	4	13
Enet(+H)	3	0*	2	1	1	2	5	11
GBRT(+H)	0	1	0	0	0	1	2*	4
RF	0	0	1	0	0	2*	2*	5
VASA	0	3*	1	0	1	2	6	13
NN1	0	1	0	0*	1	1	0	3
NN2	1*	2*	1*	3*	3*	3*	2	14
NN3	0	3	0	0	1*	1	1*	6
NN4	0	0	0	0	2	0	0	2
NN5	0*	4	0	0	0	0	0	4

- Under the CSPA totals, the best models are:

- NN4: 2 total rejections,
- NN1: 3,
- GBRT: 4,
- NN5: 4,
- RF: 5.

- By contrast, PLS has **29** total rejections and NN2 has **14**.

Economic message. Raw R^2 is not the whole story. Neural networks, especially NN4 and NN5, look more robust across macro environments than many competing methods.

5.7 Figures 3 and 4: what NN4 learns about small versus large stocks

Read.

- For monthly forecasts, the most important variables remain similar at the top across size groups, but **small stocks** receive relatively more weight on:
 - volatility and idiosyncratic risk,

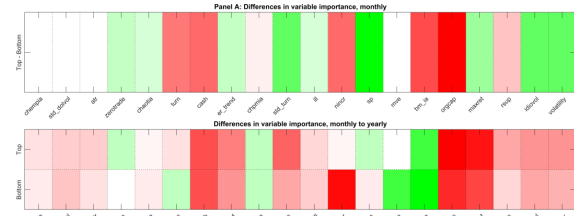


Fig. 3. Relative variable importance. This figure visualizes the changes in variable importance for the NN4 model. In Panel A, we plot the change in variable importance when moving from the top 70% to the bottom 30% of stocks for the monthly strategy. In Panel B, we plot the changes with these two groups when moving from a monthly to a yearly strategy. The red color denotes a decrease, and the green color denotes an increase in importance. The ordering of the variables corresponds to their variable importance for the whole sample of stocks at the monthly prediction horizon.

- recent extreme positive returns,
 - the China-specific trend signal,
 - liquidity-related trading signals.
- When moving from a monthly to an annual horizon, these speculative signals lose importance, and weights shift toward ownership, growth, leverage, and other slower-moving firm characteristics.

Economic message. The neural-network heterogeneity results line up with the paper’s story that retail speculation drives short-horizon predictability in small stocks, while longer-horizon predictability reflects slower-moving corporate and institutional forces.

5.8 Table 6 and Figure 5: full-sample portfolio performance

Table 6
Performance of machine learning portfolios based on the full sample (value-weighted). This table reports the out-of-sample performance measures for all machine learning models of the value-weighted long-short and long-only portfolios based on the full sample. All measures are based on 103 monthly out-of-sample returns from January 2012 to June 2020. “Avg”: average predicted monthly return (%); “Std”: the standard deviation of monthly predicted monthly returns (%); “S.R.”: annualized Sharpe ratio; “Skew”: skewness; “Kurt”: kurtosis; “Max DD”: the portfolio maximum drawdown (%); “Max 1M Loss”: the most extreme negative monthly return (%).

	“1/M” Portfolio	Machine Learning Portfolios										
		OLS-3 +H	PLS +H	LASSO +H	Einet +H	GBRT +H	RF	VASA	NN1	NN2	NN3	NN4
Long-Short												
Avg	–	1.80	3.17	3.72	3.79	3.15	2.22	4.49	5.17	4.75	5.50	5.40
Std	–	6.63	5.34	5.60	5.80	6.52	5.21	6.30	7.21	5.05	5.52	6.43
S.R.	–	0.94	2.05	2.30	2.27	1.67	1.47	2.47	2.48	3.35	3.45	2.91
Skew	–	0.58	–0.64	0.27	–0.63	–0.23	–0.76	1.21	3.53	1.35	2.49	3.44
Kurt	–	2.25	1.64	3.04	5.25	0.64	0.45	9.27	24.37	6.56	13.51	21.65
Max DD	–	45.97	17.57	15.49	29.78	24.21	16.08	16.79	13.54	7.91	5.29	6.29
Max 1M Loss	–	18.85	17.57	15.49	24.02	18.07	11.90	16.64	12.50	7.91	4.58	5.82
Long-Only												
Avg	1.56	2.45	2.74	3.37	3.35	2.59	2.22	4.04	4.23	3.84	4.36	4.50
Std	8.44	9.43	6.97	7.79	7.72	6.83	7.16	8.55	9.63	7.72	8.60	9.27
S.R.	0.64	0.89	1.42	1.49	1.50	1.31	1.07	1.64	1.52	1.72	1.76	1.76
Skew	0.26	0.49	–0.12	1.04	0.48	0.16	0.41	1.03	2.09	0.59	1.22	1.41
Kurt	1.26	1.36	1.40	4.65	2.11	2.77	1.70	4.81	10.72	2.97	5.98	6.46
Max DD	54.20	47.24	33.56	22.61	24.94	35.46	38.83	22.46	21.04	21.20	21.37	21.53
Max 1M Loss	25.56	24.66	19.66	20.95	21.42	22.54	18.49	21.22	21.04	20.28	20.34	20.16

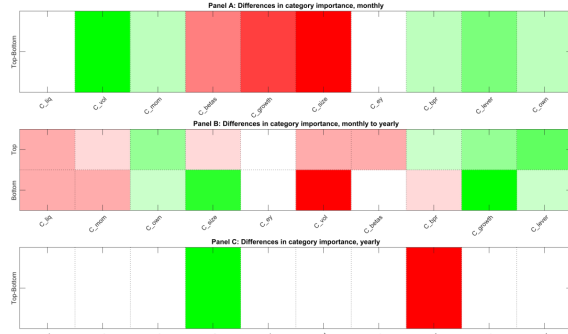


Fig. 4. Relative importance of variable categories. This figure visualizes the changes in aggregated variable importance for the NN4 model. We aggregate the variables into the categories defined in Table C4 in the Internet Appendix. Panel A shows the differences between the top 70% and the bottom 30%, and Panel B shows the corresponding changes from monthly to yearly predictions. In Panel C, we show the same graph as Panel A but for yearly predictions. The red color denotes a decrease, and the green color denotes an increase in importance. The ordering of the variables in Panel A (Panels B and C) corresponds to the median rank of the categories’ variable importance for the whole sample of stocks at the monthly (yearly) prediction horizon. Having defined these categories, we then sort them according to the median rank in monthly predictions for each category and all stocks. To analyze the differences, we look for each category at the two most important variables and how their average changes when we move from large to small stocks.

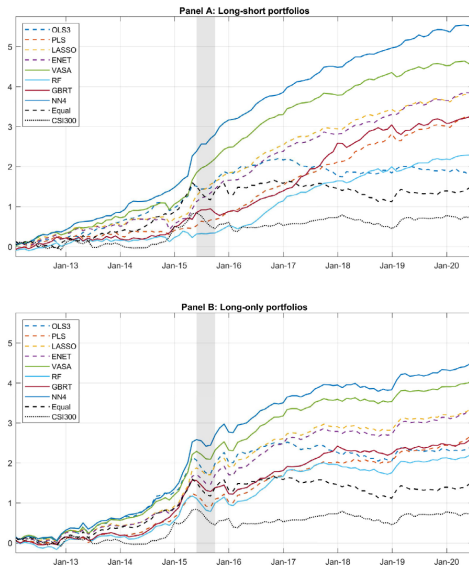


Fig. 5. Cumulative log return of machine learning portfolios (full sample). This figure shows the cumulative log returns of all portfolios and the CSI 300 market index. The shaded period corresponds to the 2015 stock market crash in China. All portfolios are constructed based on the full sample and are value weighted. In Panel A, the portfolios are based on a long-short strategy. Panel B plots the long-only portfolios.

Read.

- In the full-sample **value-weighted long-short** strategies:
 - VASA reaches a Sharpe ratio of **2.47**,
 - NN2 reaches **3.25**,
 - NN3 reaches the highest Sharpe ratio at **3.45**,
 - NN4 and NN5 reach **2.91** and **3.01**.
- In the **value-weighted long-only** strategies:
 - the 1/N benchmark Sharpe ratio is only **0.64**,
 - VASA reaches **1.64**,
 - NN2 reaches **1.72**,
 - NN3 reaches the highest at **1.76**.
- The cumulative-return figure shows that neural networks and VASA perform especially well through the **2015 crash** and do not suffer an obvious collapse during the early **COVID-19 shock** in 2020.

Economic message. The forecasts are not merely statistically significant. They also generate large and unusually resilient portfolio gains.

5.9 Tables 7 and 8: excluding small stocks and focusing on SOEs

Table 7
Performance of machine learning portfolio based on the top 70% sample (value-weighted). This table reports the out-of-sample performance measures for all machine learning models of the value-weighted long-short and long-only portfolios based on the top 70% sample. All measures are based on 103 monthly out-of-sample returns from January 2012 to June 2020. "Avg": average predicted monthly return (%); "Std": the standard deviation of monthly predicted monthly returns (%); "S.R.": annualized Sharpe ratio; "Skew": skewness; "Kurt": kurtosis; "Max DD": the portfolio maximum drawdowns (%); "Max 1M Loss": the most extreme negative monthly return (%).

	Machine Learning Portfolios												
	"1/N" Portfolio	OLS-3 +H	PLS	LASSO +H	Einet +H	GBRT +H	RF	VASA	NN1	NN2	NN3	NN4	NN5
Long-Short													
Avg	–	0.88	2.51	2.41	2.37	2.29	1.19	2.88	3.27	3.39	3.73	3.53	3.50
Std	–	5.83	5.17	4.73	5.47	6.28	5.00	4.84	4.41	4.08	4.03	4.79	4.49
S.R.	–	0.52	1.68	1.76	1.50	1.26	0.82	2.06	2.57	2.88	3.21	2.55	2.70
Skew	–	0.23	-0.41	-0.57	-1.10	-0.28	-0.88	-0.61	-0.07	0.08	0.18	0.98	0.31
Kurt	–	0.92	1.84	1.26	4.27	1.02	1.95	3.21	0.94	0.90	1.51	3.19	0.44
Max DD	–	53.80	18.29	15.22	30.78	25.69	21.90	17.01	13.54	9.50	6.35	8.59	7.52
Max 1M Loss	–	17.58	18.16	15.22	22.87	19.25	17.82	17.01	11.29	9.50	4.86	8.59	7.52
Long-Only													
Avg	1.10	1.54	1.93	2.03	1.83	1.62	1.10	2.35	2.26	2.55	2.47	2.60	2.50
Std	8.17	8.75	6.54	6.84	6.90	6.46	6.84	7.39	7.23	7.14	6.97	7.50	7.58
S.R.	0.47	0.61	1.02	1.03	0.92	0.87	0.56	1.10	1.08	1.24	1.23	1.20	1.14
Skew	0.10	0.23	-0.14	0.18	0.01	-0.37	-0.31	0.28	0.11	-0.03	-0.07	0.15	0.22
Kurt	1.32	1.10	1.68	1.82	2.27	3.85	3.41	1.68	2.24	1.68	1.67	1.97	1.99
Max DD	42.48	58.31	37.43	27.87	31.74	48.60	42.80	26.47	32.93	27.84	30.55	32.32	30.67
Max 1M Loss	26.44	24.80	20.26	22.81	23.46	25.41	26.36	22.76	23.77	22.83	22.31	23.80	23.65

Table 8
Performance of machine learning portfolios based on SOEs (value-weighted). This table reports the out-of-sample performance measures for all machine learning models of the value-weighted long-short and long-only portfolios based on SOEs. All measures are based on 103 monthly out-of-sample returns from January 2012 to June 2020. "Avg": average predicted monthly return (%); "Std": the standard deviation of monthly predicted monthly returns (%); "S.R.": annualized Sharpe ratio; "Skew": skewness; "Kurt": kurtosis; "Max DD": the portfolio maximum drawdowns (%); "Max 1M Loss": the most extreme negative monthly return (%).

	Machine Learning Portfolios												
	"1/N" Portfolio	OLS-3 +H	PLS	LASSO +H	Einet +H	GBRT +H	RF	VASA	NN1	NN2	NN3	NN4	NN5
Long-Short													
Avg	–	1.38	3.00	3.39	3.65	3.21	2.13	3.62	4.04	4.16	4.05	4.15	4.48
Std	–	4.88	4.06	3.59	4.19	3.88	3.10	4.53	3.73	3.67	3.70	3.88	3.76
S.R.	–	0.98	2.56	2.94	3.02	2.87	2.38	2.77	3.74	3.93	3.79	3.70	4.12
Skew	–	0.13	-0.57	-0.27	-0.62	-0.03	-0.76	-0.36	0.36	-0.26	-0.03	0.56	0.12
Kurt	–	0.06	0.91	0.75	2.29	-0.15	1.79	1.22	0.70	0.01	0.71	2.29	0.22
Max DD	–	34.70	14.71	10.72	16.70	8.26	9.81	13.22	7.43	6.54	10.20	10.10	9.76
Max 1M Loss	–	11.02	12.59	9.77	14.44	6.86	9.11	12.01	5.02	5.28	7.15	7.61	6.33
Long-Only													
Avg	1.13	2.00	2.42	2.62	2.86	2.67	2.17	2.87	3.04	3.16	3.11	3.18	3.35
Std	7.80	8.99	7.08	7.77	7.02	7.58	8.17	7.96	8.27	7.61	7.97	8.23	8.26
S.R.	0.50	0.77	1.19	1.17	1.25	1.22	0.92	1.25	1.27	1.44	1.35	1.34	1.41
Skew	-0.03	0.13	0.02	0.12	0.10	-0.36	-0.04	0.10	0.08	0.07	-0.04	0.23	0.18
Kurt	1.24	1.02	1.37	1.49	1.50	2.38	1.59	1.51	1.73	1.16	1.89	1.48	1.17
Max DD	54.23	52.24	30.46	26.64	24.78	34.91	41.63	25.18	28.96	23.57	25.95	25.60	24.52
Max 1M Loss	25.04	26.07	21.50	23.82	24.69	26.78	26.43	24.05	25.72	21.55	25.95	23.92	22.69

Read.

- Removing the bottom 30% of firms lowers the gains, but machine learning still dominates simple benchmarks. In the large-stock sample, NN3 still reaches a long-short Sharpe ratio of **3.21**.
- The **SOE** portfolios are surprisingly strong. In Table 8, the long-short Sharpe ratio for **NN5** reaches **4.12**, even higher than in the top-70% sample.
- For SOE long-only portfolios, neural networks also improve risk-adjusted performance relative to the 1/N benchmark and help reduce drawdowns.

Economic message. The monthly R^2 evidence says non-SOEs are more predictable at short horizons, but the portfolio analysis shows that SOE predictability can still be very economically valuable, especially for the more flexible models.

5.10 Tables 9–11: transaction costs and price limits

Table 9
Slippage for machine learning portfolios. This table reports robust average slippage (average, standard error, minimum, first quartile, third quartile) of slippage for machine learning portfolios in the testing sample, including the time-weighted average price (in bps), the value-weighted average price (in bps), and the conservative trading, robust (in bbps). The difference of TWAP, VWAP, and market capacity are detailed in the first paragraph in Section 6.4.

	OLS-5	POLS	LASSO	Lasso	GB	RF	VWAP	NN2	NN3	NN4	NN5
	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
TWAP (bps)											
Avg	2.85	2.84	3.71	2.16	2.52	6.45	1.44	2.34	3.49	3.61	3.43
Std	0.022	0.022	0.024	0.013	0.020	0.071	0.173	0.024	0.024	0.027	0.025
Skew	-3.26	-3.07	-3.34	-3.35	-3.94	-3.33	-3.47	-3.02	-3.38	-3.34	-3.27
Kurt	22.13	24.46	22.06	22.08	27.52	21.73	21.63	24.96	22.05	22.05	22.05
Min	-10.25	-8.30	-10.00	-9.34	-9.87	-7.36	-9.38	-7.35	-8.97	-7.80	-7.80
Max	16.60	22.73	15.06	23.33	26.43	36.69	24.38	23.33	23.80	23.27	25.70
VWAP (bps)											
Avg	-5.62	-7.32	-7.14	-7.72	-8.40	-8.87	-7.48	-8.38	-8.38	-7.53	-8.03
Std	0.006	0.003	0.004	0.006	0.004	0.010	0.010	0.006	0.006	0.006	0.005
Skew	1.80	1.32	1.30	1.17	1.00	1.00	1.03	1.04	1.00	1.17	1.04
Kurt	6.02	13.20	14.64	6.57	11.09	14.22	11.36	10.97	10.84	11.62	11.11
Min	-10.38	-18.24	-16.02	-20.00	-19.81	-22.23	-21.02	-18.46	-21.34	-19.78	-21.11
Max	7.04	3.13	4.05	1.38	3.34	1.26	4.47	3.73	3.74	4.00	4.34
Market Capacity											
Avg	3.08	3.07	3.18	1.85	3.15	5.00	0.97	2.40	3.00	3.13	2.75
Std	0.40	0.001	0.10	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Skew	1.39	1.00	1.00	-4.37	-3.78	-3.88	-4.00	-3.78	-3.87	-3.87	-3.87
Kurt	24.42	26.08	27.20	27.11	25.78	25.13	27.08	26.63	26.43	26.01	26.01
Min	-10.71	-7.92	-12.29	-11.38	-9.40	-8.85	-10.38	-8.83	-9.58	-7.53	-8.33
Max	30.21	32.00	34.01	35.46	36.43	36.09	37.62	34.94	37.02	35.00	35.00

Table 10
Portfolio performance including transaction costs (value-weighted). This table reports the impact of transaction costs on the monthly return (in %) and the associated Sharpe ratio of the portfolios averages based on different machine learning algorithm.

Monthly return											
Sharpe ratio											
Long-Short											
Transaction costs	0 bps	20 bps	40 bps	60 bps	80 bps	0 bps	20 bps	40 bps	60 bps	80 bps	0 bps
OLS-5 (H)	7.28	7.04	6.95	6.30	5.48	3.83	3.87	3.88	3.40	3.11	3.11
OLS-5 (LH)	1.80	1.86	1.53	1.39	1.23	0.84	0.87	0.80	0.71	0.66	0.66
POLS	3.17	3.00	2.82	2.63	2.41	1.65	1.64	1.64	1.44	1.33	1.42
LASSO (H)	5.72	5.48	5.25	2.89	2.75	2.36	2.15	1.89	1.84	1.69	1.69
LASSO (LH)	1.19	1.00	1.00	0.91	0.80	0.67	0.66	0.66	0.66	0.66	0.66
GB (H)	2.22	2.01	1.80	1.59	1.38	1.47	1.35	1.20	1.06	0.92	0.92
VWAP	6.49	6.17	6.00	3.84	3.65	3.47	3.33	3.11	2.99	2.99	2.99
RF	5.17	4.91	4.65	4.39	4.12	2.48	2.36	2.23	2.10	1.97	1.97
NN2	6.70	6.50	6.24	3.89	3.73	3.28	3.09	2.91	2.73	2.55	2.55
NN3	5.98	5.84	5.68	4.71	4.40	3.40	3.28	3.12	2.98	2.79	2.79
NN4	5.40	5.14	4.87	4.61	4.35	2.91	2.76	2.62	2.48	2.34	2.34
NN5	5.51	5.25	4.97	4.69	4.43	3.01	2.85	2.70	2.55	2.39	2.39
Long-Only											
Transaction costs	0 bps	20 bps	40 bps	60 bps	80 bps	0 bps	20 bps	40 bps	60 bps	80 bps	0 bps
OLS-5 (H)	3.03	2.87	2.72	2.55	2.41	1.34	1.28	1.21	1.14	1.07	1.07
OLS-5 (LH)	2.04	2.15	2.24	2.17	2.07	0.80	0.80	0.80	0.76	0.76	0.76
POLS	2.74	2.64	2.55	2.44	2.37	1.42	1.37	1.33	1.28	1.23	1.23
LASSO (H)	1.97	1.81	1.70	1.57	1.45	1.00	0.94	0.89	0.84	0.79	0.79
GB (H)	3.35	3.21	3.07	2.92	2.78	1.30	1.24	1.17	1.11	1.04	1.04
VWAP (H)	2.57	2.47	2.35	2.21	2.10	1.13	1.07	1.01	0.95	0.89	0.89
RF	2.22	2.10	1.99	1.89	1.79	1.04	1.00	0.94	0.87	0.80	0.80
NN2	4.20	4.08	3.94	3.80	3.66	1.01	0.97	0.94	0.89	0.84	0.84
NN3	3.84	3.70	3.58	3.43	3.29	1.72	1.66	1.60	1.54	1.48	1.48
NN4	4.36	4.21	4.09	3.94	3.80	1.76	1.70	1.64	1.59	1.53	1.53
NN5	4.00	4.06	4.21	4.07	3.93	1.68	1.63	1.57	1.52	1.46	1.46

Table 11
Impacts of machine learning portfolios. This table reports the out-of-sample performance measures for all machine learning models of the equally-weighted long-only and long-only portfolios with trading costs, i.e., excluding costs of price limits. All measures are based on 30 monthly out-of-sample returns from January 2012 to June 2020. "Avg." average predicted monthly return (in %); "SA." annualized Sharpe ratio; "Nontradeable" fraction of stocks that are not tradeable.

Machine learning portfolios											
	OLS-5	POLS	LASSO	GB	RF	VWAP	NN2	NN3	NN4	NN5	NN5
	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Long-only											
Avg	2.24	2.07	4.05	4.20	3.81	4.48	4.30	4.45	4.74	4.91	4.91
SA	0.80	0.68	1.64	1.64	1.54	1.64	1.64	1.64	1.77	1.79	1.79
Tradeable											
Avg	2.23	2.05	3.74	3.91	3.51	3.21	4.08	4.18	4.42	4.59	4.53
SA	0.84	0.75	1.47	1.49	1.44	1.31	1.57	1.55	1.68	1.70	1.65
Nontradeable	0.1	0.3	0.8	0.4	0.1	0.7	0.8	0.7	0.7	0.3	0.8

Read.

- Average total slippage from TWAP and VWAP calculations is on the order of roughly **10 bps** after combining the buy and sell sides, which motivates conservative round-trip cost scenarios.
- Even at **80 bps** round-trip transaction cost, the NN4 long-short Sharpe ratio falls only from **2.91** to **2.34**, and the NN4 long-only Sharpe ratio falls from **1.68** to **1.46**.
- Daily price limits do not overturn the results. For NN4, the long-only Sharpe ratio declines only from **1.78** to **1.70** after excluding non-tradeable stocks at the limits.

Economic message. The paper makes a serious implementation effort. The machine-learning gains survive realistic Chinese-market frictions rather than disappearing once the portfolios become tradable.

6 Short summary

- The paper asks whether modern machine-learning return-prediction methods work in the Chinese stock market and whether the answers differ from the U.S.
- Using 1,160 predictors built from stock characteristics, macro variables, interactions, and industry dummies, the authors show that machine learning substantially improves out-of-sample forecasts.
- The biggest monthly gains come from **small, retail-heavy**, and **non-SOE** stocks; the biggest annual gains come from **large stocks** and **SOEs**.
- The most important predictors in China are **liquidity-based** signals, not the momentum ranking emphasized in the U.S. evidence.
- Neural networks are the most robust models overall, and the resulting portfolio strategies remain economically meaningful after transaction costs and daily price limits.

7 Conclusion

7.1 Core conclusion

The paper shows that machine-learning return prediction works very well in the Chinese stock market. Relative to both simple linear benchmarks and much of the U.S. evidence, out-of-sample predictability is stronger, especially in market segments most exposed to retail trading and speculative behavior. Neural networks emerge as the most robust model class once the authors combine raw out-of-sample R^2 , conditional predictive-ability tests, and portfolio performance.

7.2 Economic intuition

The Chinese market appears to contain **two distinct sources of predictability**. At short horizons, predictability is strongest where retail investors are likely to matter most, namely in small and small-shareholder stocks. This is why liquidity, abnormal turnover, volatility, and sentiment-related signals are so central. At longer horizons, predictability shifts toward large firms and SOEs, which the paper interprets as consistent with the importance of slower-moving government signaling and institutional forces.

7.3 Takeaway

For studying asset pricing with machine learning, market structure matters. The Chinese stock market does not just give a larger version of the U.S. result. It changes which signals dominate, where predictability is concentrated, and how much of that predictability survives in realistic trading strategies. That is the paper's main contribution.